Integrated Pan-Cancer Multi-Omics Analysis Reveals *TMEM45A* Associated with DNA Hypomethylation, Immune Exclusion, and Drug Response

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Abstract

Background: Transmembrane Protein 45A (*TMEM45A*) has been implicated in hypoxia tolerance and chemoresistance in several cancers; however, its molecular landscape, epigenetic regulation, and influence on the tumor immune microenvironment (TME) have not been systematically characterized across human malignancies.

Methods: This study performed an integrated pan-cancer multi-omics analysis across 33 malignancies using transcriptomic, proteomic, epigenomic, pharmacogenomic, and single-cell datasets from TCGA, GTEx, and public resources. Transcriptomic expression, clinicopathology, protein abundance, genomic alterations, DNA methylation, immune infiltration, drug sensitivity, and functional networks were systematically evaluated. Prognostic significance was determined via univariate Cox regression and Kaplan–Meier survival analyses.

Results: TMEM45A was significantly overexpressed across multiple solid tumors (including BRCA, GBM, HNSC, KIRC, LUSC), where elevated expression correlated with advanced pathological stage and poorer overall (OS) and disease-free survival (DFS). Upregulation was driven by promoter and probe-specific (cg07907506) DNA hypomethylation. Low-frequency copy-number amplifications (up to 8% in LUSC) and recurrent DUF716 domain mutations (V236I) showed limited prognostic value. TMEM45A expression inversely correlated with $CD8^+$ T-cell infiltration, but positively associated with cancer-associated fibroblasts (CAFs), M2 macrophages, stromal/immune scores, and immune checkpoints (PDCD1, CTLA4, HAVCR2), indicating an immune-excluded phenotype. Pharmacogenomic analysis revealed consistent inverse sensitivity to docetaxel and a top positive correlation with vorinostat across GDSC and CTRP datasets.

Conclusion: Integrated network analysis identified TMEM189 as associated with TMEM45A based on convergent protein interaction and transcriptional co-expression analyses. Collectively, these findings establish a comprehensive pan-cancer framework for understanding TMEM45A biology and connect the TMEM45A–TMEM189 network as a molecular axis for future mechanistic and translational investigation.

Keywords: *TMEM45A*; Pan-cancer analysis; Multi-omics; DNA hypomethylation; Tumor microenvironment; Cancer prognosis; Pharmacogenomics; *TMEM189*.

Introduction

Despite remarkable advances in precision oncology, therapeutic resistance and tumor heterogeneity remain major barriers to improved long-term outcomes across many human malignancies(1–3). Global estimates reported approximately 20 million new cancer cases and 9.7 million deaths in 2022, with Asia accounting for nearly half of all cases and over half of fatalities, underscoring the region's disproportionate burden (4,5). Despite substantial advances in oncology and systemic therapies, advanced and metastatic cancers remain difficult to manage owing to biological heterogeneity and therapeutic resistance (6,7). Consequently, identifying molecular biomarkers that improve diagnosis, prognostic assessment, and therapeutic stratification remains a major challenge in precision oncology (8,9).

Immune checkpoint blockade has revolutionized therapy across multiple malignancies, yet durable clinical responses remain restricted to a minority of patients (10,11). This variability reflects TME complexity, where dynamic cell-cell interactions dictate immune surveillance and therapeutic response. Identifying TME-reflective biomarkers is thus crucial for prognosis and precision oncology (12–14).

Expanding public multi-omics datasets have revolutionized cancer systems biology, enabling comprehensive genomic, transcriptomic, proteomic, and epigenetic characterizations across diverse human malignancies (15–17). Pan-cancer multi-omics integration provides a systematic framework to identify shared molecular vulnerabilities, prioritize targets, and elucidate conserved oncogenic pathways beyond tissue boundaries. Within this framework, TMEM45A is a understudied seven-pass transmembrane protein localized to the trans-Golgi network and endoplasmic reticulum(18–20). TMEM45A has been implicated in cellular stress adaptation and the maintenance of cell survival under hypoxic conditions (21–23).

Despite accumulating evidence supporting the oncogenic role of TMEM45A, significant knowledge gaps remain regarding its genomic alterations, epigenetic regulation, immune-related functions, and predictive value for treatment response across diverse malignancies (24,25). Previous studies have linked TMEM45A to epithelial–mesenchymal transition (EMT), tumor progression, and drug resistance; however, these findings remain largely context-dependent and occasionally inconsistent across human cancers (26). Furthermore, TMEM45A's influence on immune cell infiltration and checkpoint expression remains uncharacterized, and no study has comprehensively integrated multi-omics analyses to evaluate its pan-cancer role. To address these knowledge gaps, we developed an integrated pan-cancer multi-omics framework combining transcriptomic, proteomic, epigenetic, single-cell, immunological, and pharmacogenomic datasets to systematically characterize TMEM45A and explore its functional relationship with TMEM189 across human malignancies (27,28).

To address these knowledge gaps, we performed an integrated pan-cancer multi-omics analysis of TMEM45A across 33 human malignancies. We systematically characterized its transcriptomic/proteomic expression, clinicopathology, prognosis, genomic alterations, DNA methylation, TME interactions, single-cell dynamics, drug sensitivity, and functional networks using large-scale public datasets. Integrative analyses identified TMEM189 as a co-expressed functional partner and revealed predicted sensitivity to docetaxel alongside resistance to the HDAC inhibitor vorinostat, offering testable hypotheses for future translational studies.

Materials and Methods

Data Sources and Study Design

This study analyzed public multi-omics datasets primarily from The Cancer Genome Atlas (TCGA), with external validation where appropriate. The overall analytical workflow is summarized in **Figure 1**.

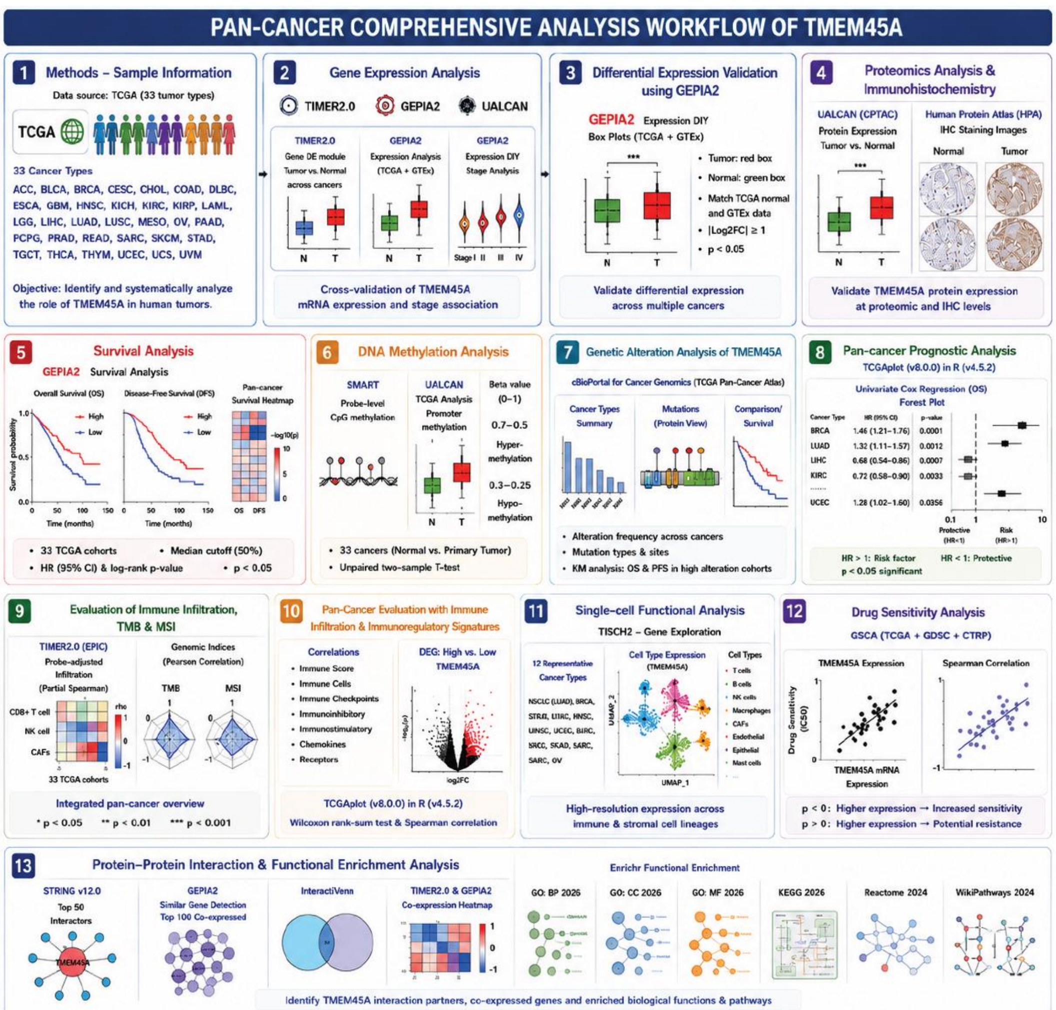

Figure 1: The *TMEM45A* pan-cancer comprehensive analysis workflow. The multi-omics approach used to comprehensively discover and describe the oncogenic, prognostic, and immunological effects of TMEM45A across 33 human tumor types is shown in this schematic pipeline.

Cell Line Authentication and Mycoplasma Testing

As this study utilized no experimental cell lines or laboratory assays, cell line authentication and mycoplasma testing were not applicable.

Gene Expression Analysis

TMEM45A transcriptomic expression across TCGA malignancies was evaluated using the Gene_DE module of TIMER2.0(29). Differential expression between tumor and normal tissues was independently validated using the GEPIA2 Expression DIY module, integrating TCGA and GTEx datases to cmpensate for limited TCGA normal samples ($P < 0.05$, $|\log_2 FC| \geq 1$)(30). Correlations between TMEM45A expression and pathological stage were assessed via GEPIA2 and cross-validated using UALCAN(31). Differential TMEM45A expression between tumor and GTEx normal tissues was validated using the GEPIA2 Expression DIY module to compensate for limited TCGA normal controls, with significance set at $P < 0.05$ and $|\log_2 FC| \geq 1$(30).

Proteomics Analysis and Immunohistochemistry

TMEM45A protein abundance was evaluated across CPTAC datasets via UALCAN (31). Representative immunohistochemical (IHC) staining from the Human Protein Atlas (HPA) validated protein expression patterns in paired tumor and normal tissues(32).

Survival Analysis

The prognostic significance of TMEM45A across 33 TCGA malignancies was evaluated via GEPIA2. Overall survival (OS) and disease-free survival (DFS) were assessed using median expression to stratify patients into high- and low-expression groups. Survival differences were evaluated by the log-rank test, reporting hazard ratios (HRs) and 95% confidence intervals (CIs) ($P < 0.05$ considered significant) (30).

DNA Methylation Analysis

Probe-level CpG methylation of TMEM45A across TCGA cohorts was evaluated using SMART (33). Promoter methylation across 33 TCGA malignancies was analyzed in UALCAN, comparing primary tumors against paired normal controls (34). Methylation was quantified as β-values (0 = unmethylated, 1 = fully methylated; hypermethylation: 0.50–0.70; hypomethylation: 0.25–0.30), with group differences evaluated by unpaired two-sample t-test (31).

Genetic Alteration Analysis of *TMEM45A*

TMEM45A genetic alterations were evaluated across the TCGA PanCancer Atlas dataset via cBioPortal(35). Alteration frequencies, mutation types, and cross-cancer distributions were assessed using the Cancer Types Summary module, while amino acid substitution patterns across protein domains were examined via the Mutations module. Associations between TMEM45A alterations and clinical outcomes—specifically overall survival (OS) and progression-free survival (PFS)—were evaluated by Kaplan–Meier analyses using the Comparison/Survival module (35).

Pan-cancer Prognostic Analysis

The prognostic value of TMEM45A across TCGA cohorts was evaluated using univariate Cox proportional hazards regression via the *TCGAplot* package (v8.0.0) in R (v4.5.2), reporting hazard ratios (HRs) and 95% confidence intervals (CIs) for overall survival (OS) as forest plots(36). An HR > 1 indicated elevated mortality risk, while an HR < 1 indicated reduced risk (two-sided $P < 0.05$ considered significant).

Immune Infiltration, Tumor Mutation Burden, and Microsatellite Instability Analysis

Immune cell infiltration ($CD8^+$ T cells, NK cells, and CAFs) was evaluated across TCGA cohorts using TIMER2.0 with the EPIC algorithm(29,37). Confounding non-tumor cellular components were controlled via purity-adjusted partial Spearman correlation. Correlations between TMEM45A expression and tumor mutation burden (TMB) or microsatellite instability (MSI) were evaluated by Pearson correlation. Data processing and visualization were performed using ggplot2, dplyr, and *TCGAplot* in R (v4.5.2)(36,38). Significance was set at $P < 0.05$ ($* P < 0.05$, $** P < 0.01$, $*** P < 0.001$).

Immune Regulatory Signature Analysis

Associations between TMEM45A expression and immune regulatory signatures—including immune scores, immune checkpoints, immunoinhibitors, immunostimulators, chemokines, and chemokine receptors—were evaluated across TCGA cohorts. Differential expression between high- and low-TMEM45A groups was evaluated by Wilcoxon rank-sum test using TCGAplot (v8.0.0) in R (v4.5.2), while gene correlations were assessed using Spearman's rank correlation(36).

Single-Cell Transcriptomic Analysis

Single-cell TMEM45A expression was evaluated using the Gene module of TISCH2 across 12 cancer types: LUAD (NSCLC datasets), BRCA, STAD, LIHC, HNSC, UCEC, BLCA, KIRC, SKCM, PAAD, SARC, and OV (39). Representative datasets with validated cell-type annotations were analyzed to characterize the cellular distribution of TMEM45A across annotated tumor, immune, and stromal cell populations (39).

Drug Sensitivity Analysis

The association between TMEM45A expression and anticancer drug sensitivity was evaluated using the Gene Set Cancer Analysis (GSCA) platform (40). GSCA integrates TCGA transcriptomic profiles with pharmacogenomic datasets from the Genomics of Drug Sensitivity in Cancer (GDSC) and Cancer Therapeutics Response Portal (CTRP) (41,42). Spearman correlation analysis assessed relationships between TMEM45A expression and half-maximal inhibitory concentration (IC_{50}) values. A positive correlation (higher IC_{50}) indicated reduced drug sensitivity, whereas a negative correlation (lower IC_{50}) indicated greater predicted sensitivity.

Protein–Protein Interaction and Functional Enrichment Analysis

A TMEM45A-centered protein–protein interaction (PPI) network was constructed in STRING (v12.0) using all evidence sources and a minimum interaction score of 0.150 to maximize coverage for exploratory analysis(43). The top 50 interacting proteins ranked by STRING confidence score were retained for downstream analyses. The top 50 interacting proteins were intersected with the top 100 positively co-expressed genes identified from integrated TCGA/GTEx data via GEPIA2(30) using InteractiVenn(44).Co-expression patterns were validated using TIMER2.0 (Gene_Corr) and GEPIA2 (Correlation Analysis)(29,30). Functional enrichment analyses—

including Gene Ontology (BP, CC, MF) and pathway databases (KEGG, Reactome, WikiPathways)—were performed using Enrichr(45–48).

Analysis of statistics

Differential expression between tumor and normal tissues was evaluated using the Wilcoxon rank-sum test in TIMER2.0 and Student's t-test or one-way ANOVA in UALCAN. Unless specified by the platform, correlation analyses utilized Spearman's rank correlation coefficient. A two-sided $P < 0.05$was considered statistically significant.

Results

Expression Level of *TMEM45A*

TMEM45A exhibited significant cancer-type-specific dysregulation across TCGA cohorts compared to corresponding normal tissues. Specifically, TMEM45A was significantly downregulated in CHOL, KICH, LIHC, PRAD, THCA, and UCEC, but upregulated in BRCA, GBM, HNSC, KIRC, and LUSC **(Figure 2A)**. Additionally, expression was significantly higher in primary cutaneous melanoma than in metastatic SKCM lesions. These patterns were independently validated in UALCAN **(Figure 2B)**. Pathological stage analysis revealed significant associations with TMEM45A expression in BRCA, KICH, KIRC, and LUSC ($P < 0.05$; **Figure 2C**), whereas CHOL, HNSC, LIHC, SKCM, THCA, and UCEC showed no stage dependence **(Supplementary Figure 1)**. Stage-wise evaluation was omitted for GBM and PRAD due to incomplete clinical annotation.

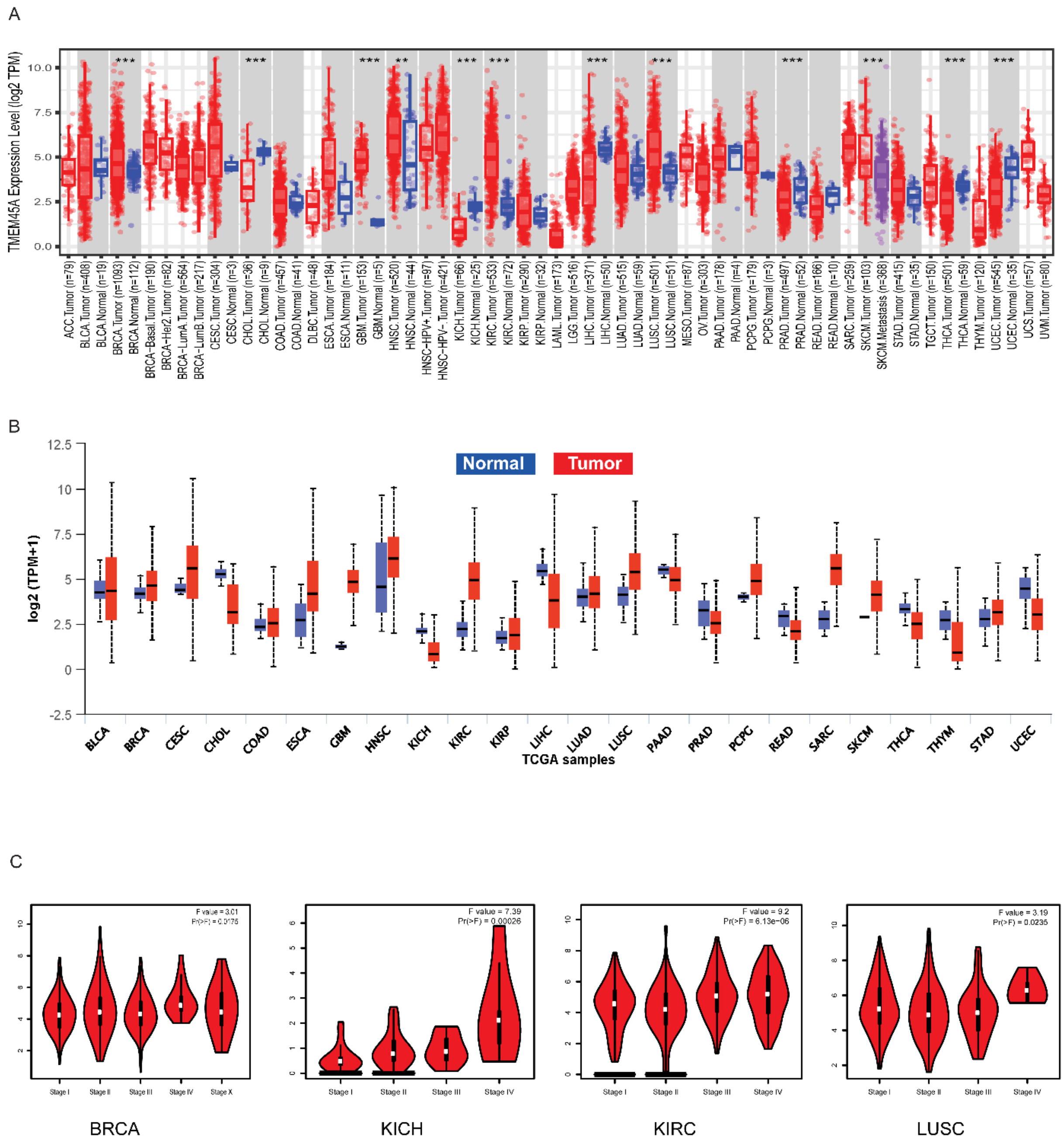

Figure 2: *TMEM45A*'s pan-cancer transcriptome expression profile and pathological stage connection. **(A)** TIMER2.0 profiles human pan-cancer *TMEM45A* expression levels across several TCGA cohorts, with asterisks (*$p < 0.05$, $p < 0.01$, ***$p < 0.001$) denoting statistical significance. **(B)** Using UALCAN, box plots of tumor (red) and normal (blue) tissues from available TCGA datasets were validated for differential *TMEM45A* expression. **(C)** Using GEPIA2, pathological stage-wise expression analysis of *TMEM45A* in BRCA, KICH, KIRC, and LUSC cohorts (F-statistics and p-values indicated).

Validation of *TMEM45A* Overexpression via GEPIA2

Integrated TCGA–GTEx analysis in GEPIA2 confirmed cancer-type-specific dysregulation of TMEM45A. TMEM45A was significantly elevated in DLBC, GBM, HNSC, KIRC, LGG, LUSC, PAAD, and SARC ($P < 0.05$), validating TIMER2.0 findings **(Figure 3A)**. Conversely, expression was significantly decreased in CHOL, KICH, LAML, LIHC, READ, SKCM, TGCT, THCA, and UCEC **(Figure 3B)**, with no significant differences observed in remaining malignancies **(Supplementary Figures S2A, S2B).**

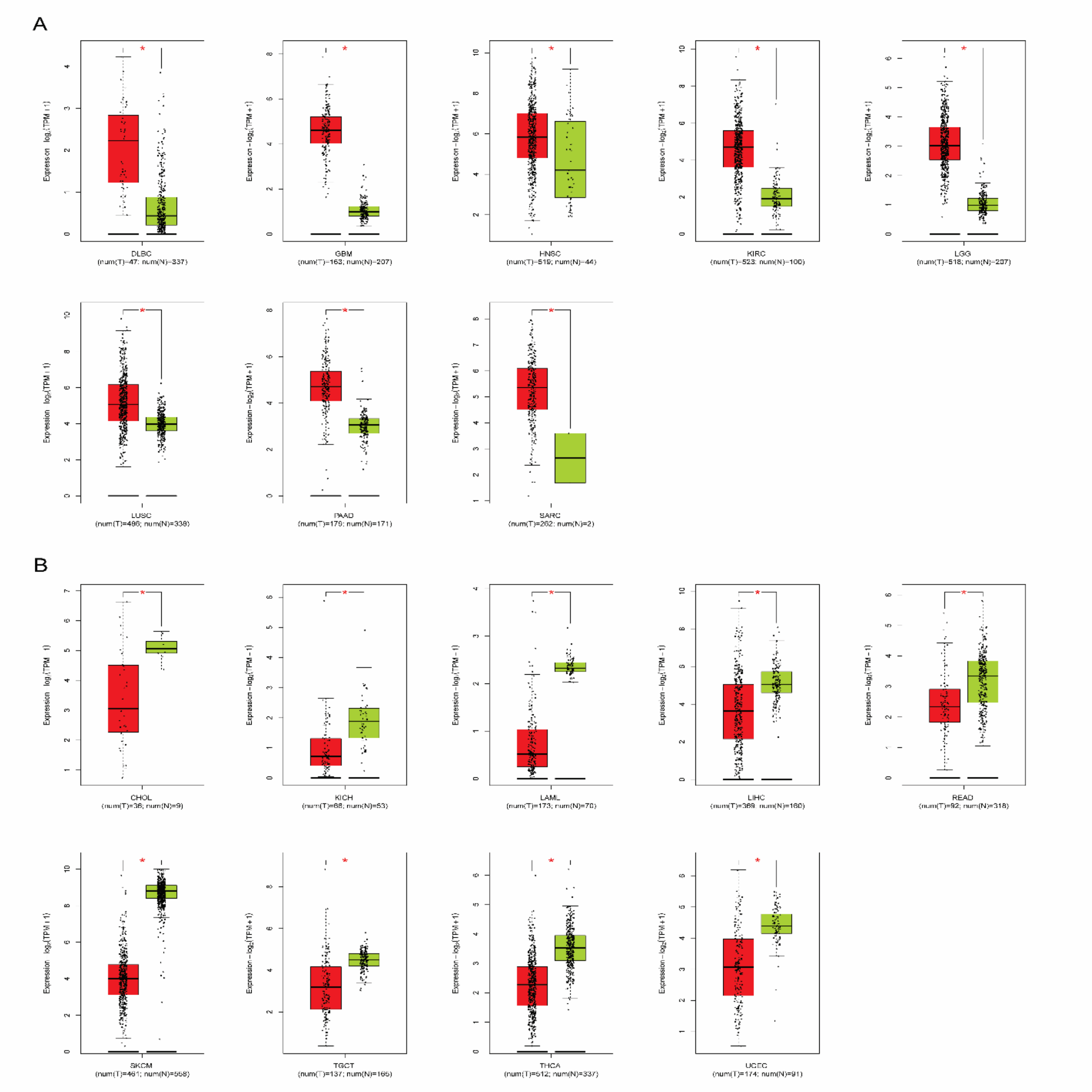

Figure 3: *TMEM45A* expression differences between TCGA and GTEx cohorts are validated using

GEPIA2. *TMEM45A* transcriptome expression differences between primary tumor (red) and normal control (green) tissues are validated by detailed box plots. **(A)** Cohorts such as DLBC, GBM, HNSC, KIRC, LGG, LUSC, PAAD, and SARC showed substantial transcriptome up-regulation ($p < 0.05$). **(B)** Cohorts such as CHOL, KICH, LAML, LIHC, READ, SKCM, TGCT, THCA, and UCEC showed substantial transcriptome down-regulation ($p < 0.05$). Below each corresponding graphic are the sample sizes for the tumor (num(T)) and normal (num(N)) cohorts.

Proteomic Analysis of *TMEM45A*

CPTAC proteomic datasets via UALCAN, available for four TCGA cohorts, demonstrated significantly higher TMEM45A protein abundance in BRCA ($P < 1 \times 10^{-12}$), HNSC ($P = 3.53 \times 10^{-6}$), and LIHC ($P = 7.33 \times 10^{-4}$) versus normal tissues, while LUSC showed no significant difference ($P = 0.297$ **(Supplementary Figure 3)**. Immunohistochemical (IHC) profiling using the HPA062101 antibody (Human Protein Atlas) confirmed malignant TMEM45A protein localization **(Figure 4, Table 1)**. SKCM displayed strong cytoplasmic and nuclear immunoreactivity versus normal skin; THCA exhibited moderate staining in malignant follicular cells with unmeasurable staining in normal thyroid; and LIHC showed positive staining in malignant hepatocyte cords. These proteomic and histological data broadly concord with transcriptomic observations.

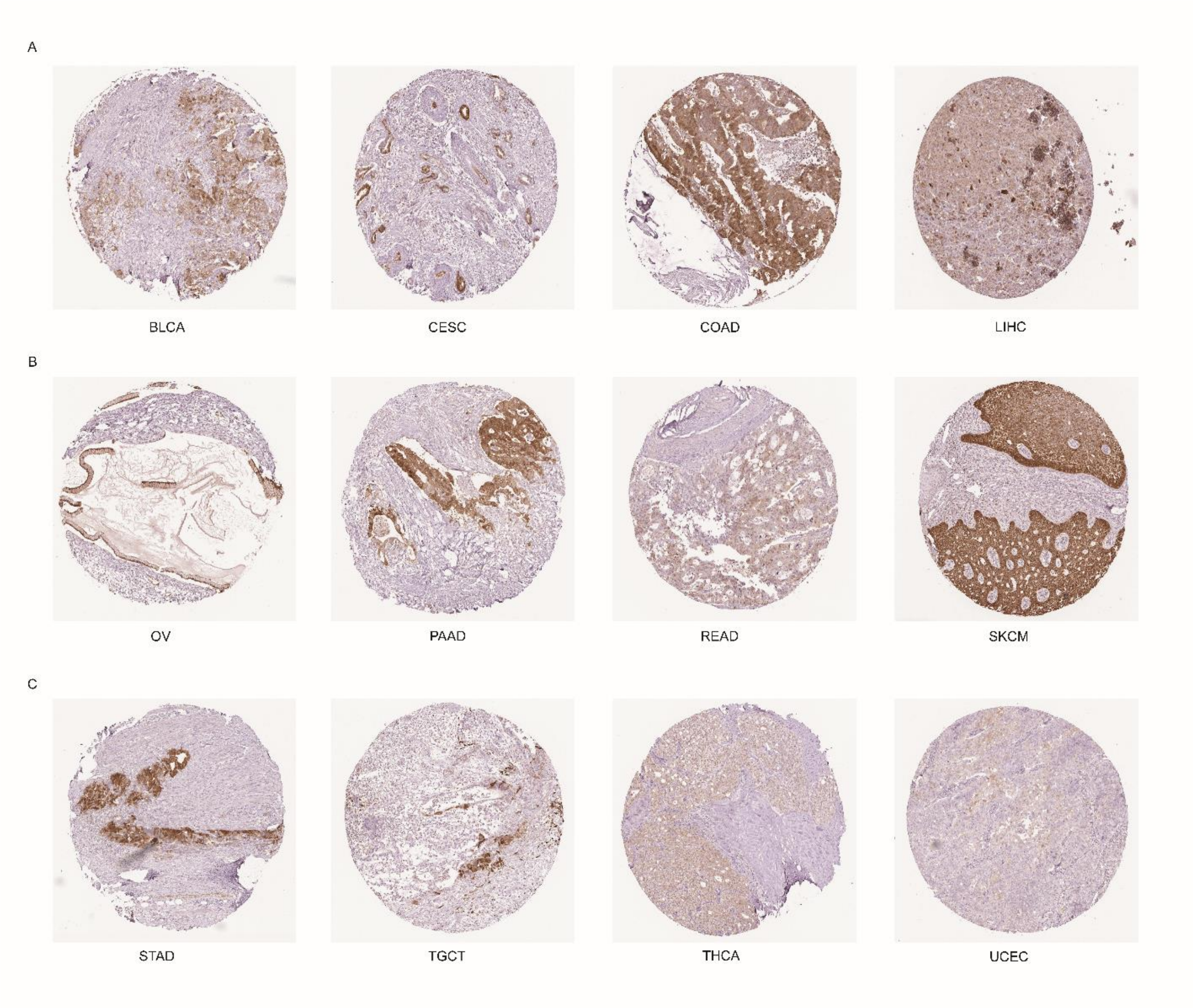

Figure 4: *TMEM45A* protein expression in several tumor tissues is validated by immunohistochemistry. Using the HPA062101 antibody, representative immunohistochemistry (IHC) tissue cores were acquired from the Human Protein Atlas (HPA) database. Twelve typical cancer types, such as BLCA, CESC, COAD, LIHC, OV, PAAD, READ, SKCM, STAD, TGCT, THCA, and UCEC, are represented by distinct microarrays that show varying protein expression and staining intensity **(A–C).** Variations in nuclear and cytoplasmic staining density throughout the malignant cellular compartments are indicative of immunoreactivity.

Table 1. Immunohistochemical protein expression profile of *TMEM45A* across different human cancers based on the Human Protein Atlas database.

Cancer	Expression (Staining level)	Intensity	Quantity	Location
BLCA	Medium	Moderate	75% - 25%	Cytoplasmic/membranous
CESC	High	Strong	75% - 25%	Cytoplasmic/membranous nuclear
COAD	High	Strong	75% - 25%	Cytoplasmic/membranous nuclear
LIHC	Low	Weak	75% - 25%	Cytoplasmic/membranous nuclear
OV	High	Strong	75% - 25%	Cytoplasmic/membranous nuclear
PAAD	High	Strong	75% - 25%	Cytoplasmic/membranous nuclear
READ	Low	Moderate	<25%	Nuclear
SKCM	High	Strong	>75%	Cytoplasmic/membranous nuclear
STAD	High	Strong	75% - 25%	Cytoplasmic/membranous nuclear
TGCT	Medium	Strong	<25%	Cytoplasmic/membranous
THCA	Medium	Moderate	75% - 25%	Cytoplasmic/membranous
UCEC	Not detected	Weak	<25%	Nuclear

BLCA: Bladder Urothelial Carcinoma; **CESC:** Cervical Squamous Cell Carcinoma and Endocervical Adenocarcinoma; **COAD:** Colon Adenocarcinoma; **LIHC:** Liver Hepatocellular Carcinoma; **OV:** Ovarian Serous Cystadenocarcinoma; **PAAD:** Pancreatic Adenocarcinoma; **READ:** Rectum Adenocarcinoma; **SKCM:** Skin Cutaneous Melanoma; **STAD:** Stomach Adenocarcinoma; **TGCT:** Testicular Germ Cell Tumors; **THCA:** Thyroid Carcinoma; **UCEC:** Uterine Corpus Endometrial Carcinoma. Immunohistochemical staining data were generated using the HPA062101 antibody obtained from the Human Protein Atlas database. Detailed information for individuals listed in **supplementary table S1**. The remaining cancer types showed no detectable TMEM45A protein expression in the Human Protein Atlas database, with antibody staining classified as not detected, with negative intensity, no quantity, and no specific location.

Prognostic Value of *TMEM45A* in Human Cancers

Elevated TMEM45A expression was associated with unfavorable clinical outcomes across multiple TCGA cohorts **(Figures 5A, 5B).** Survival heatmaps and Kaplan–Meier analyses revealed that high TMEM45A significantly correlated with poorer overall survival (OS) in KIRC ($P = 0.014$), KIRP

($P = 0.041$), LIHC ($P = 0.049$), STAD ($P = 0.003$), and THCA ($P = 0.0087$) **(Figures 5A, 5C).** Additionally, high expression predicted significantly shorter disease-free survival (DFS) in KIRC ($P = 1 \times 10^{-4}$), OV ($P = 0.019$), and STAD ($P = 0.027$) **(Figures 5B, 5D)**. Concurrently, KIRC and STAD demonstrated associations across both OS and DFS.

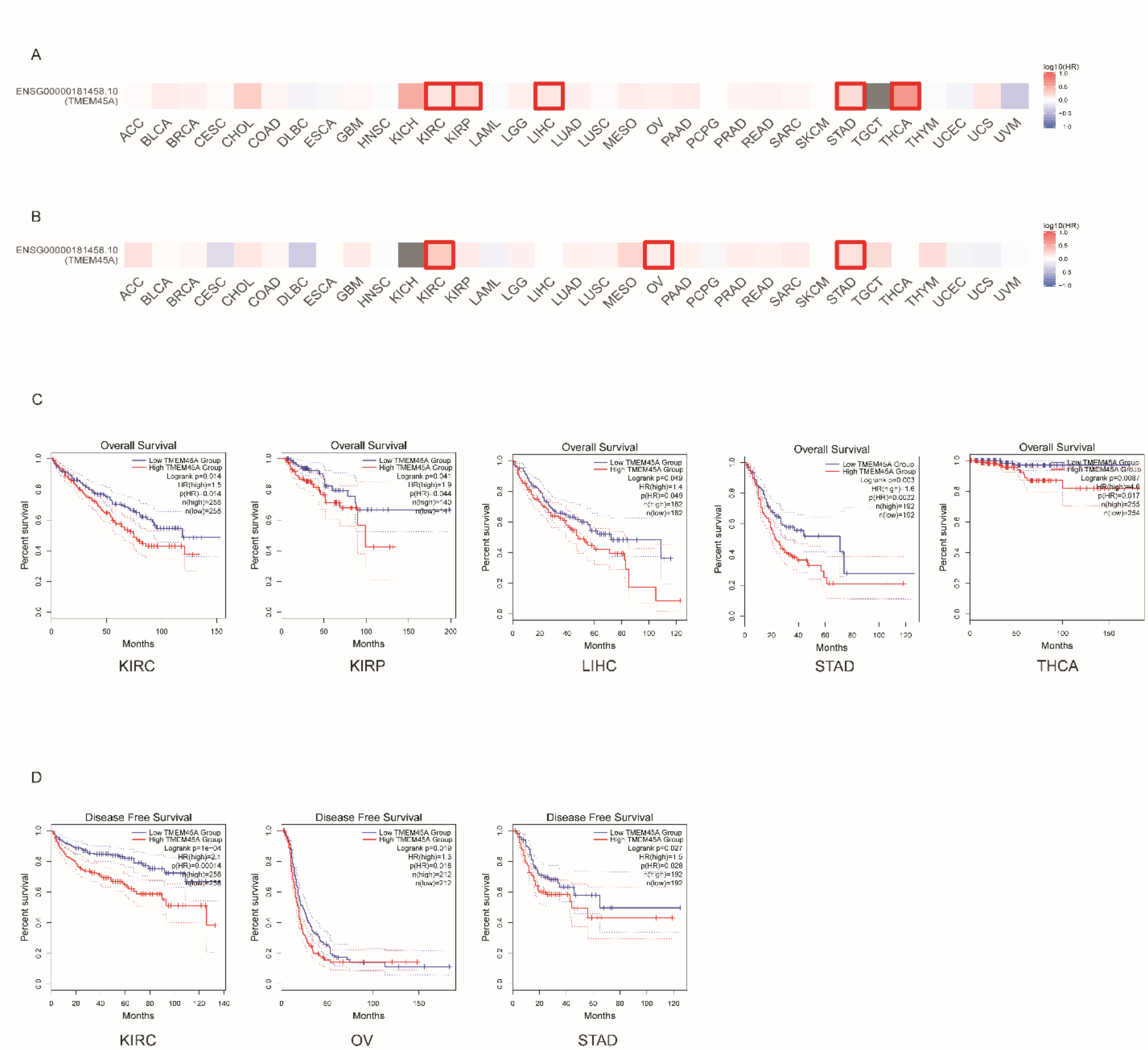

Figure 5: Survival study and pan-cancer prognostic classification of *TMEM45A* expression in TCGA cohorts. **(A, B)** Pan-cancer survival heatmaps showing the statistical relationship between *TMEM45A* expression. **(A)** Overall Survival (OS) and **(B)** Disease-Free Survival (DFS) in 33 human cancers. Cohorts with statistically significant survival risk are shown by red frames. To assess **(C)** OS across KIRC, KIRP, LIHC, STAD, and THCA, and **(D)** DFS across KIRC, OV, and STAD, Kaplan-Meier survival curves stratified cohorts into high (red) and low (blue) *TMEM45A* expression groups. Each corresponding curve has annotations for log-rank p-values, hazard ratios (HR), and sample sizes (n).

Methylation Analysis

DNA methylation profiling revealed widespread TMEM45A hypomethylation across TCGA cohorts **(Figure 6A).** Significant hypomethylation at the N-shore CpG probe cg07907506 was observed in BRCA, CHOL, KIRC, LIHC, LUAD, LUSC, PRAD, and UCEC versus normal controls ($P < 0.05$; **Figure 6B**; locus details in **Supplementary Figure S4**), supported by CpG-aggregated promoter analysis **(Figure 6C).**

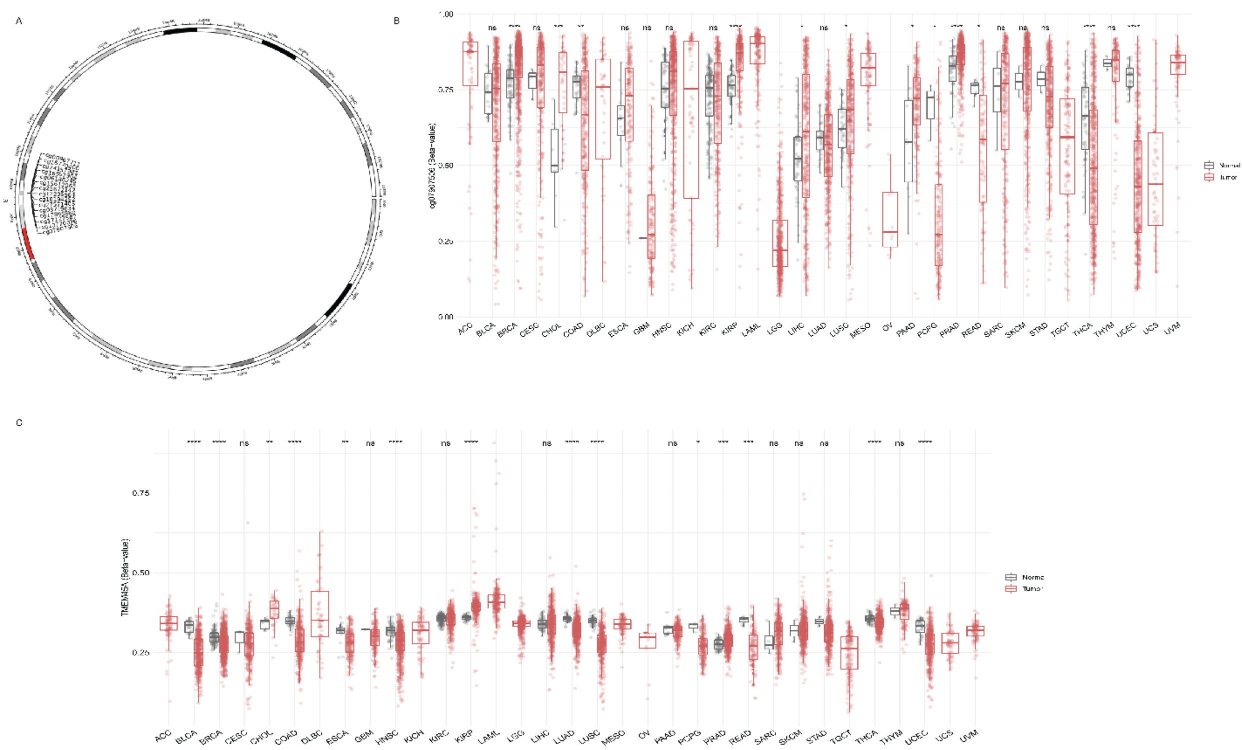

Figure 6: TCGA cohorts' DNA methylation profiles and the *TMEM45A* locus epigenetic environment. **(A)** A circular chromosomal ideogram from the SMART database that shows the **TMEM45A** gene locus's structural landscape and genomic location. **(B)** Box plots employing the particular N-shore probe cg07907506 across many TCGA cohorts that show the differences in DNA methylation levels (Beta-values) between primary tumors (red) and normal controls (blue). **(C)** CpG-aggregated methylation analysis showing the total methylation status of the TMEM45A promoter in several human malignancies. Asterisks (*p < 0.05, **p < 0.01, ***p < 0.001) show statistical significance, whereas ns indicates non-significant differences.

UALCAN analysis confirmed significant promoter hypomethylation ($\beta < 0.3$, $P < 0.05$) in BRCA, CESC, CHOL, KIRP, LUSC, PAAD, PCPG, PRAD, READ, THCA, and UCEC **(Figure 7).** No significant differences were detected in ESCA, GBM, HNSC, KIRC, LIHC, LUAD, SARC, STAD, or THYM, while normal control data were absent for ACC, DLBC, KICH, LAML, LGG, MESO, SKCM, TGCT, UCS, and

UVM. Cross-platform comparison between SMART and UALCAN confirmed concordant hypomethylation in BRCA, CHOL, and UCEC, despite platform-specific variations in GBM, KICH, and SKCM. Collectively, dual-platform analyses confirm the promoter hypomethylation mechanism of TMEM45A upregulation in select malignancies.

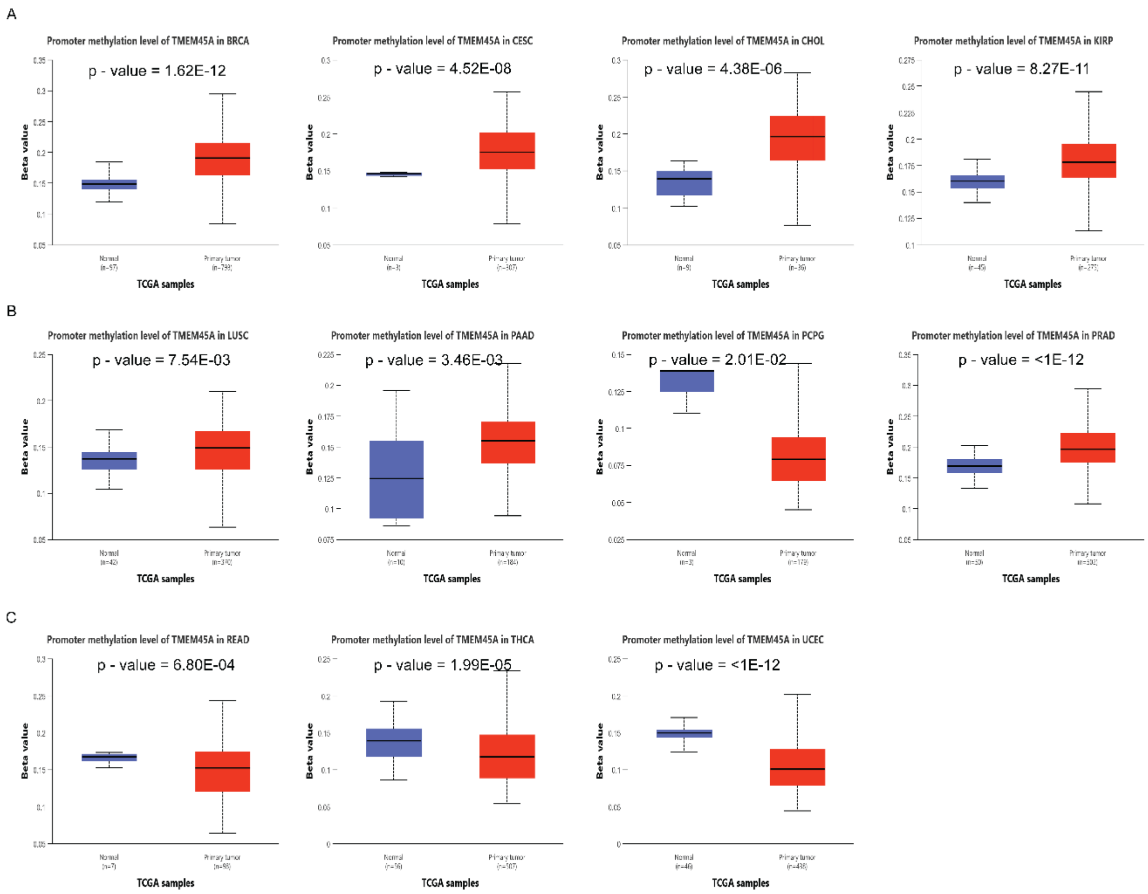

Figure 7: UALCAN is used to validate the pan-cancer *TMEM45A* promoter methylation levels. Box plots showing substantial promoter hypomethylation (beta-values) across assessed TCGA cohorts between primary tumor tissues (red) and normal controls (blue). **(A)** Notable patterns of promoter hypomethylation in CHOL, KIRP, CESC, and BRCA. **(B)** Promoter methylation patterns and epigenetic differences in LUSC, PAAD, PCPG, and PRAD. **(C)** THCA, UCEC, and READ validation curves. Each panel is marked with sample sizes (n) and precise statistical p-values.

Genetic Alteration Analysis of *TMEM45A*

TMEM45A displayed low yet cancer-type-specific genomic alteration frequencies across TCGA cohorts, with gene amplification representing the predominant alteration type. Highest alteration frequencies were observed in LUSC (~8%), CESC (~5.5%), ESCA (~4.3%), HNSC (~3.8%), UCEC (~3.8%), OV (~3.0%), SARC (~2.4%), and UCS (~2.0%) **(Figure 8A).**

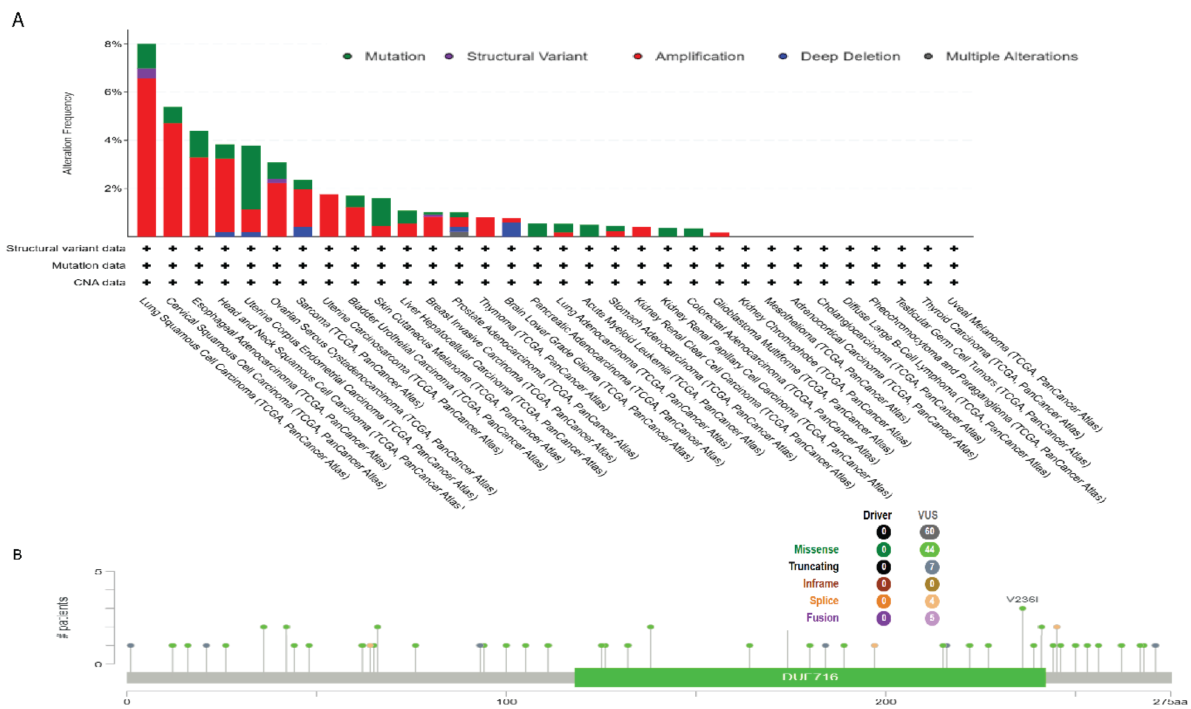

Figure 8: *TMEM45A*'s mutational landscape and pan-cancer genomic alteration profile. **(A)** *TMEM45A* alteration frequency and distribution types (amplification, deep deletion, missense mutation, structural variation, or several alterations) across TCGA cohorts are displayed in a bar chart. **(B)** Schematic figure showing the topological distribution of somatic mutations over the 275 amino acid length of the *TMEM45A* protein, with the recurrent V236I site highlighted and variant types (missense, truncating, splice, and fusion) clustered inside the DUF716 domain.

Mutational profiling across the 275-amino-acid protein identified 60 variants—comprising 44 missense, seven truncating, four splice-site, and five fusion events—with a mutation hotspot, including the recurrent V236I substitution, localized within the DUF716 domain **(Figure 8B)**. Kaplan–Meier survival analyses across the eight top-altered cohorts revealed no significant associations between TMEM45A alterations and either OS or PFS ($P > 0.05$), indicating genomic alterations do not independently predict clinical outcome in these malignancies.

Univariate Cox Regression Analysis of TMEM45A

Univariate Cox proportional hazards regression analysis demonstrated that elevated TMEM45A expression significantly correlated with poorer overall survival ($HR > 1$) across multiple TCGA cohorts (Figure 9). Specifically, elevated TMEM45A expression was significantly associated with an increased risk of death (HR > 1) in CHOL (HR = 1.596, 95% CI: 1.090–2.337, *P* = 0.016), KICH (HR = 6.293, 95% CI: 2.207–17.948, *P* = 0.001), KIRC (HR = 1.146, 95% CI: 1.033–1.270, *P* = 0.010), LGG (HR = 1.432, 95% CI: 1.113–1.843, *P* = 0.005), LIHC (HR = 1.146, 95% CI: 1.013–1.296, *P* = 0.030), LUAD (HR = 1.164, 95% CI: 1.030–1.317, *P* = 0.015), STAD (HR = 1.221, 95% CI: 1.007–1.482, *P* = 0.042), and THCA (HR = 5.749, 95% CI: 2.555–12.936, *P* < 0.001). These findings underscore the broad prognostic utility of TMEM45A across human malignancies.

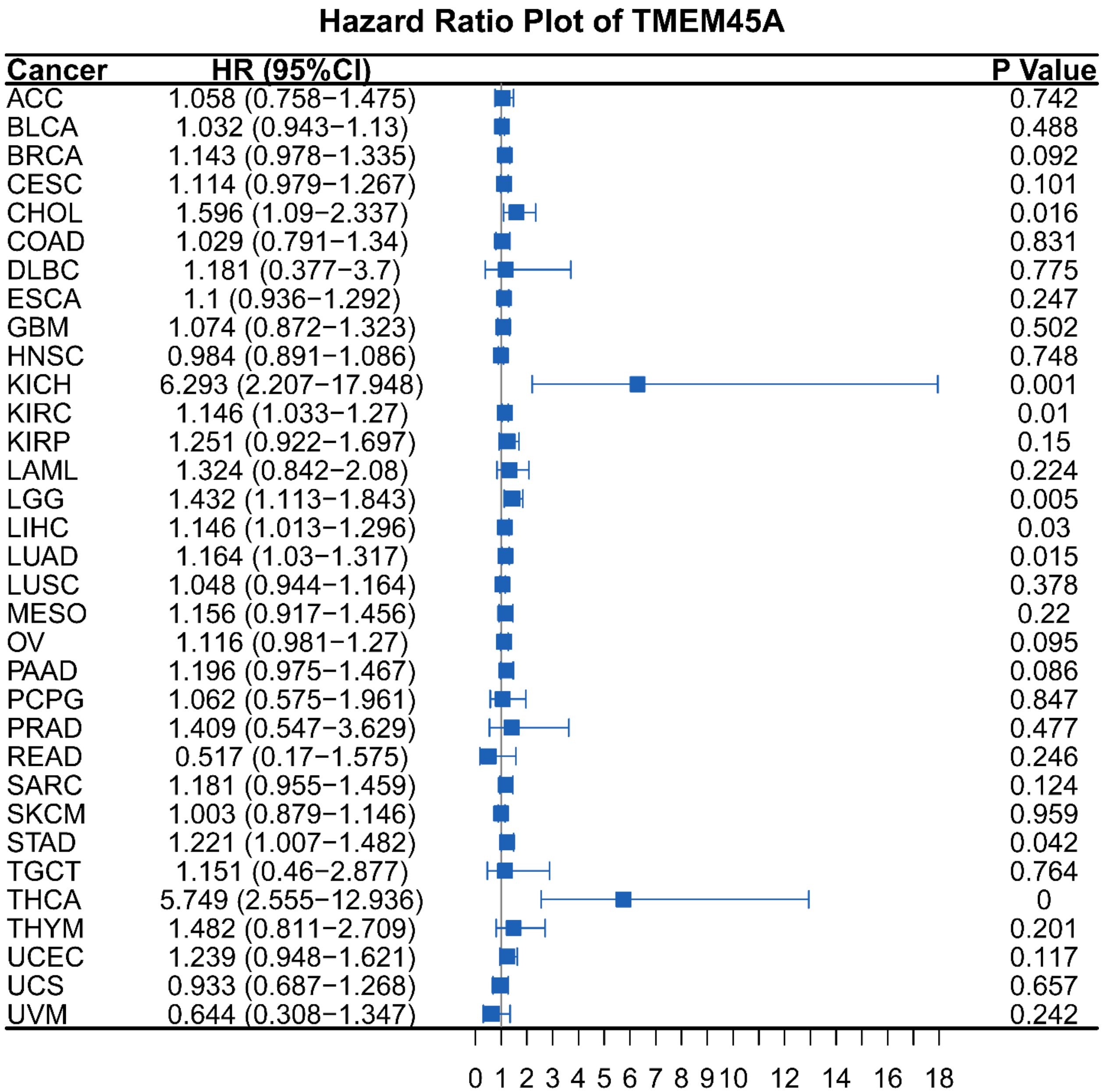

Figure 9: *TMEM45A* expression and overall survival across TCGA cohorts in a univariate Cox regression forest plot. The predictive influence of *TMEM45A* transcriptome expression across 33 human cancer types is evaluated using a forest plot that shows the hazard ratios (HR), 95% confidence intervals (CI), and associated log-rank p-values ($P < 0.05$). *TMEM45A* is identified as an adverse survival risk factor by alignments to the right of the vertical reference line (HR = 1). The panel arrangement is labeled with precise statistical matrices and cohort notations.

Immune Infiltration Landscape of *TMEM45A*

TMEM45A expression significantly correlated with immune cell infiltration, tumor mutation burden (TMB), and microsatellite instability (MSI) across multiple TCGA cohorts **(Figure 10; Supplementary Table S1).**

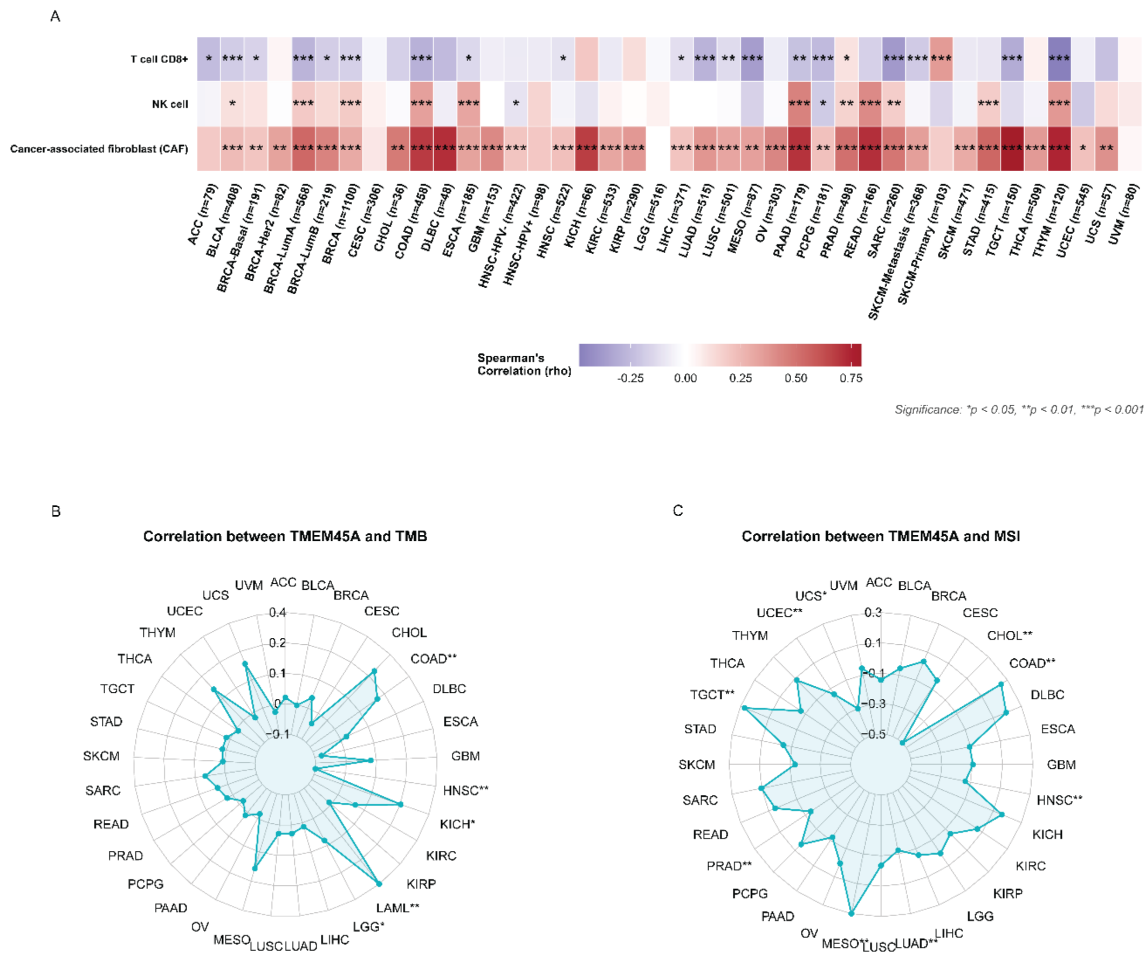

Figure 10: TCGA pan-cancer cohorts integrated multi-omics landscape of *TMEM45A* expression with immune infiltration, tumor mutation burden (TMB), and microsatellite instability (MSI). **(A)** Heatmap showing the purity-adjusted partial Spearman's rank correlation coefficients (rho) calculated using the EPIC method between *TMEM45A* mRNA expression and the infiltration abundance of CD8+ T cells, natural killer (NK) cells, and cancer-associated fibroblasts (CAFs). The correlation strength is shown by the color gradient, where blue tiles indicate negative correlations and red tiles indicate positive correlations. **(B)** Radar plot displaying the Pearson correlation coefficients between TMB and *TMEM45A* expression in various cancers. **(C)** A radar map showing the Pearson correlation coefficients between MSI status and *TMEM45A* expression in different

cohorts. Asterisk markers (*p < 0.05, **p < 0.01, and ***p < 0.001) are used to indicate statistically significant relationships across all panels. **Supplementary Table S1** contains complete quantitative value charts and precise p-values.

EPIC deconvolution revealed that TMEM45A expression predominantly correlated positively with cancer-associated fibroblast (CAF) infiltration, but negatively with $CD8^+$ T-cell infiltration **(Figure 10A)**. $CD8^+$ T-cell associations were significant in ACC, BLCA, BRCA, COAD, ESCA, HNSC, LIHC, LUAD, LUSC, MESO, PAAD, PCPG, PRAD, SARC, SKCM (primary/metastatic), TGCT, and THYM. NK-cell associations were heterogeneous, with significant correlations in BLCA, BRCA, COAD, ESCA, HNSC-HPV^-, PAAD, PCPG, PRAD, READ, SARC, STAD, and THYM.

Pearson correlation identified significant TMB associations **(Figure 10B)**—positive in LAML ($P < 0.01$), KICH ($P < 0.05$), COAD ($P < 0.01$), and LGG ($P < 0.05$), and negative in HNSC ($P < 0.01$). Additionally, TMEM45A is significantly associated with MSI **(Figure 10C)**—positively in MESO, COAD, LUAD, TGCT, and LUSC ($P < 0.01$), and negatively in CHOL ($P < 0.01$), PRAD ($P < 0.01$), HNSC ($P < 0.01$), UCEC ($P < 0.01$), and UCS ($P < 0.05$). These findings link TMEM45A expression to tumor immunobiology and microenvironmental remodeling.

Immune Regulatory Landscape of TMEM45A

TMEM45A expression is significantly associated with tumor immune microenvironment features, including ESTIMATE scores, immune cell composition, immune checkpoints, and immunoinhibitory signatures across TCGA cohorts **(Figure 11).**

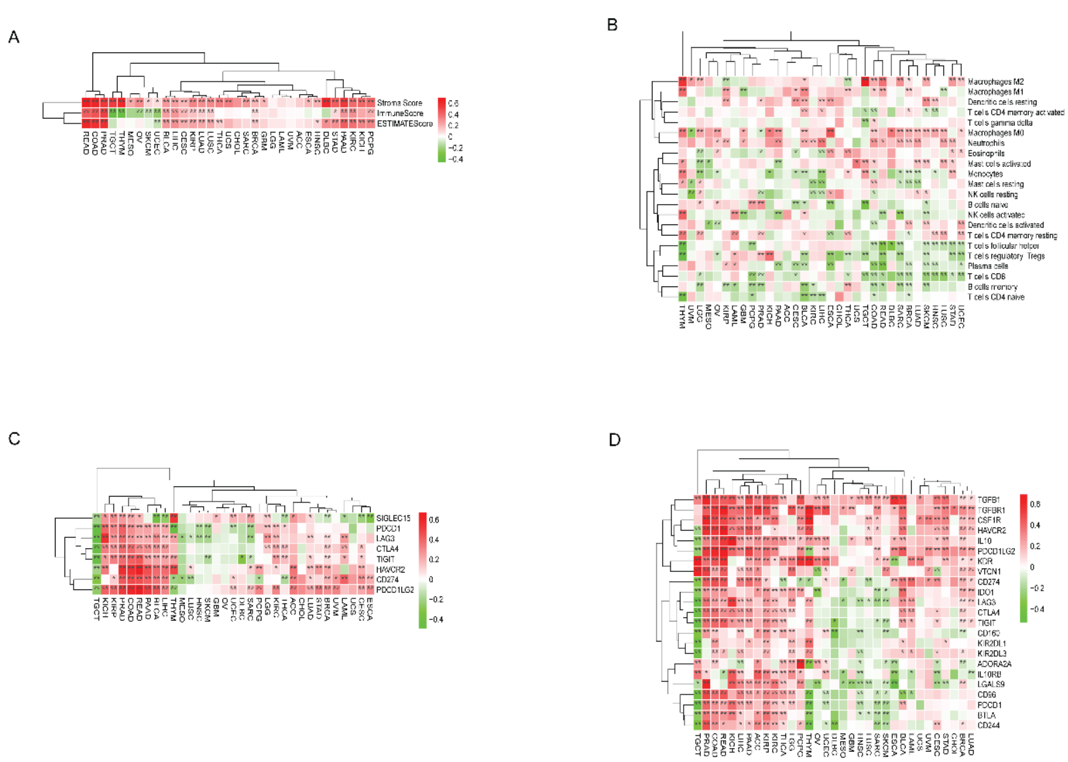

Figure 11: *TMEM45A* pan-cancer profile and its relationship to immunoinhibitory networks, cellular infiltration, treatment checkpoints, and global microenvironment dynamics. **(A)** Partial Spearman correlation coefficients between *TMEM45A* expression and immunological score across TCGA cohorts are shown in a global microenvironment immunescore heatmap. **(B)** 22 Infiltration correlation heatmap showing the connection between immune cells and *TMEM45A*. **(C)** Co-expression heatmap showing the transcriptional association between immunological checkpoints and *TMEM45A*. **(D)** A correlation heatmap showing the transcriptional co-regulation patterns of immune regulatory genes (immunoinhibitory) and *TMEM45A*. The color intensity scale indicates statistically significant correlation values (green indicates negative correlation, red indicates positive correlation).

ESTIMATE profiling revealed predominantly positive correlations between TMEM45A expression and StromalScore, ImmuneScore, and ESTIMATEScore, with significant concurrence across all three metrics in BLCA, BRCA, CESC, COAD, KIRC, KIRP, LIHC, LUAD, LUSC, PAAD, PCPG, PRAD, READ, and STAD **(Figure 11A)**. CIBERSORT analysis across 22 immune cell types showed TMEM45A expression positively correlated with macrophage subsets (M0, M1, M2), neutrophils, and resting memory $CD4^+$ T cells, but negatively correlated with $CD8^+$ T cells, regulatory T cells (Tregs), T follicular helper cells, activated NK cells, plasma cells, and memory B cells **(Figure 11B)** indicating myeloid enrichment alongside reduced lymphocyte infiltration.

Analysis of immune checkpoints demonstrated widespread positive correlations with $SIGLEC15$, $PDCD1$ (PD-1), $LAG3$, $CTLA4$, $TIGIT$, $HAVCR2$ (TIM-3), $CD274$ (PD-L1), and $PDCD1LG2$ (PD-L2), particularly in BLCA, COAD, KICH, KIRP, LIHC, PAAD, PRAD, READ, and THYM, alongside ACC, BRCA, CESC, CHOL, ESCA, LAML, LUAD, STAD, and UVM **(Figure 11C)**. Conversely, TGCT exhibited inverse correlations. Similarly, TMEM45A positively correlated with key immunoinhibitors—including $TGFB1$, $TGFBR1$, $IL10$, $CSF1R$, $HAVCR2$, KDR (VEGFR2), and $IDO1$—most prominently in COAD, KICH, PRAD, and READ, while TGCT showed inverse trends. Collectively, TMEM45A expression is linked to an immunosuppressive tumor microenvironment across human cancers.

Immunostimulatory and Chemokine Signatures Associated with TMEM45A

TMEM45A expression exhibited widespread associations with immunostimulatory molecules, chemokine signaling networks, and differential expression patterns across TCGA cohorts (Figure 12).

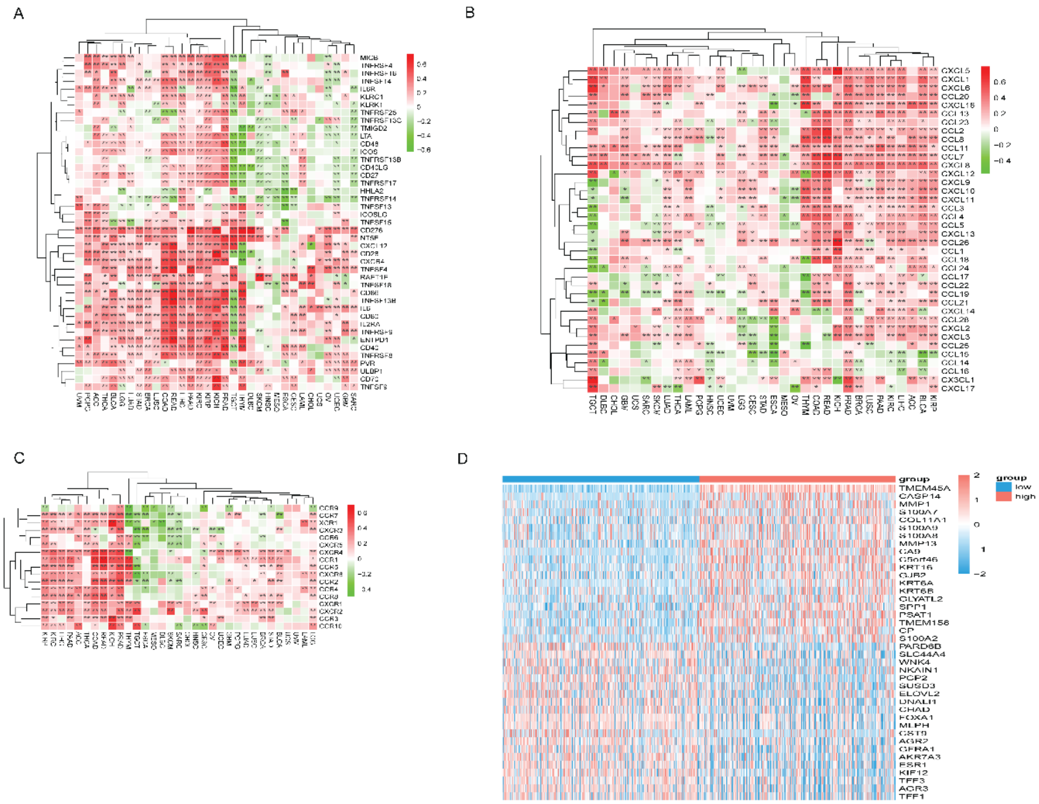

Figure 12: *TMEM45A*'s impact on chemokine signaling, immunostimulators, and pan-cancer differential expression is described. **(A)** Immunostimulatory Profiles: This heatmap shows the relationship between *TMEM45A* and a variety of immunostimulatory molecules across TCGA types, such as CD276, NT5E, and PVR. **(B)** Chemokine Ligands: *TMEM45A* and the main chemokine ligands involved in immune cell recruitment are transcriptionally correlated. **(C)** Chemokine Receptors: Heatmap displaying the co-expression of *TMEM45A* with chemokine receptors, emphasizing the alignment with particular axes like CXCL12-CXCR4. **(D)** Differential Expression Analysis: A pan-cancer heatmap and plot comparing *TMEM45A* mRNA levels between tumor (T) and adjacent normal (N) tissues. Wilcoxon rank-sum tests are used to identify significant upregulation (red) or downregulation (green).

Co-expression analysis identified positive correlations between TMEM45A and immunostimulatory genes—including $CD276$, $NT5E$, PVR, $MICB$, and $CD40$—predominantly in GBM, HNSC, OV, SARC, and UCEC **(Figure 12A)**. Conversely, inverse correlations with $CD27$, $CD28$, and $TNFRSF17$ occurred in KICH, KIRP, PRAD, and TGCT. Chemokine ligand analysis revealed positive correlations between TMEM45A and $CXCL12$, $CCL2$, and $CCL5$ in LAML, LGG, and UVM, alongside negative correlations in PRAD and TGCT **(Figure 12B)**. TMEM45A expression also correlated positively with chemokine receptors $CCR2$, $CCR5$, $CXCR4$, and $CXCR6$ in BRCA, HNSC, LUAD, and STAD **(Figure 12C)**. Finally, differential expression profiling reconfirmed significant TMEM45A upregulation across multiple tumor cohorts alongside marked downregulation in KICH and PRAD **(Figure 12D)**. Collectively, these data link TMEM45A to pan-cancer chemokine dynamics and immune microenvironment remodeling.

Single-Cell Transcriptomic Analysis of TMEM45A

Single-cell RNA sequencing across 12 representative tumor microenvironments, BLCA, BRCA, HNSC, KIRC, LIHC, NSCLC, OV, PAAD, SARC, SKCM, STAD, and UCEC—revealed cell type–specific TMEM45A distribution (Supplementary Figure S5). TMEM45A was predominantly expressed in malignant epithelial populations in BRCA, NSCLC, and STAD. Expression was further enriched in stromal lineages (fibroblasts and endothelial cells) and myeloid compartments (macrophages and monocytes), with minimal detection in T and B lymphoid populations. These single-cell patterns corroborate bulk transcriptomic deconvolution, confirming TMEM45A enrichment within stromal and myeloid tumor microenvironments.

Pharmacogenomic Associations of TMEM45A Across Anticancer Agents

TMEM45A expression significantly correlated with therapeutic sensitivity across the GDSC and CTRP datasets **(Figure 13; Supplementary Tables S2, S3)**. In GDSC, TMEM45A expression correlated positively with IC_{50} values for most associated compounds—including BHG712, BMS345541, I-BET-762, LAQ824, Methotrexate, Navitoclax, NPK76-II-72-1, PHA-793887, TPCA-1, and Vorinostat—indicating reduced drug sensitivity at higher expression levels **(Figure 13A)**. Inverse correlations were limited to Docetaxel, Bleomycin ($50\,\mu$M), and ZSTK474. In CTRP, TMEM45A similarly showed predominantly positive correlations with drug response **(Figure 13B)**, with highly significant associations ($-\log_{10}(\text{FDR}) \geq 9.0$) for belinostat, ciclopirox, CR-1-31B, gemcitabine, panobinostat, sotrastaurin, and vorinostat. Cross-dataset concordance for Vorinostat consistently links elevated TMEM45A expression to reduced HDAC inhibitor sensitivity, positioning TMEM45A as a potential pharmacogenomic marker for therapeutic stratification.

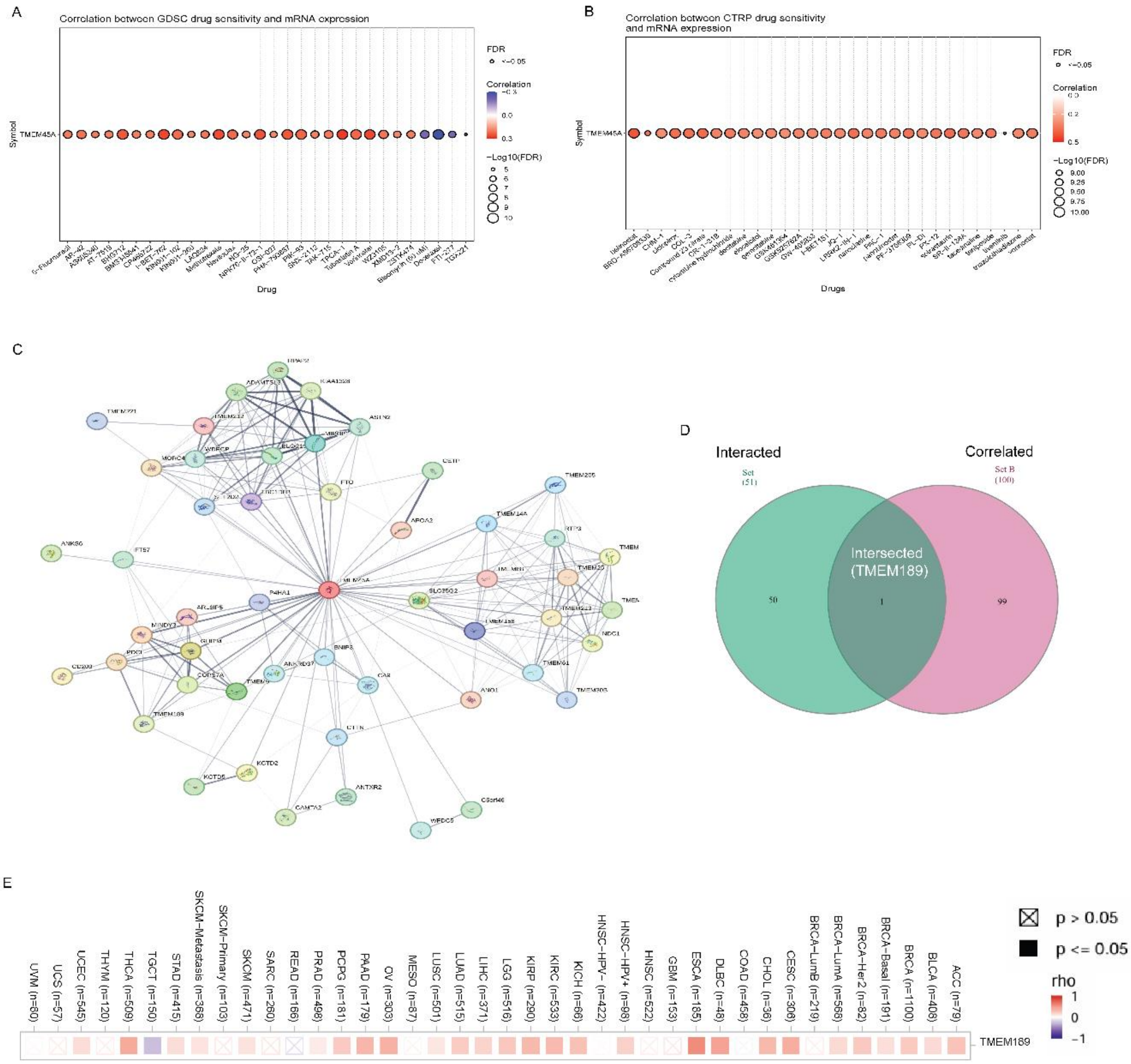

Figure 13: *TMEM45A* molecular interaction network study and drug sensitivity profiling. Bubble plots showing the relationship between drug sensitivity (IC50) and *TMEM45A* mRNA expression in the GDSC and CTRP datasets **(A, B)**. Drug resistance is shown by positive correlations (red), whereas drug sensitivity is indicated by negative correlations (blue). The statistical significance is shown by the size of the bubbles (-log10 FDR). *Docetaxel* sensitivity and *Vorinostat* resistance are strongly correlated with high *TMEM45A* expression. **(C)** The STRING database was used to create the protein-protein interaction (PPI) network of *TMEM45A*, which displays functional relationships with proteins including *TMEM189, TMEM25*, and *APOA2*. **(D)** Venn diagram illustrating the intersection of the top 100 co-expressed genes in GEPIA2 and the STRING PPI network (51 proteins). The only shared functional partner was found to be *TMEM189*. **(E)** Using TIMER2.0, a pan-cancer co-expression heatmap confirms the positive connection between

TMEM45A and *TMEM189* mRNA expression across several TCGA cohorts. Statistically significant positive associations ($p < 0.05$) are indicated by red intensity. **Supplementary Tables S2–S4** include detailed data for these studies.

TMEM45A-Centered Protein–Protein Interaction Network

STRING analysis predicted a TMEM45A-centered protein–protein interaction (PPI) network enriched for transmembrane and functionally associated proteins **(Figure 13C)**. Interacting transmembrane partners included TMEM189, TMEM25, TMEM14A, TMEM9, TMEM30B, TMEM158, TMEM221, and TMEM205, with TMEM189 demonstrating the highest combined interaction score. Additional predicted network nodes included APOA2, CETP, FTO, NDC1, IPPK, and APMAP. Collectively, this predicted network delineates functional candidate partners to guide future mechanistic studies of TMEM45A.

Integrated Network Analysis Identifies TMEM189 as a Candidate Functional Partner of TMEM45A

Integrated cross-platform analysis identified TMEM189 as the sole convergent candidate gene across 151 evaluated targets, intersecting 51 predicted STRING interaction partners and 100 GEPIA2 co-expressed genes **(Figure 13D; Supplementary Table S4)**. This alignment across predicted protein-protein interaction and transcriptomic co-expression profiles prioritizes TMEM189 as the primary functional candidate partner of TMEM45A for downstream mechanistic investigation.

Pan-Cancer Validation of TMEM45A–TMEM189 Co-expression

Pan-cancer correlation analysis demonstrated a predominantly positive co-expression pattern between TMEM45A and TMEM189 across the majority of TCGA tumor cohorts **(Figure 13E)**. Significant positive correlations were confirmed in BRCA, CESC, CHOL, DLBC, ESCA, KICH, KIRC, KIRP, LGG, LIHC, LUAD, OV, PAAD, PCPG, and THCA, whereas TGCT exhibited an inverse correlation. These transcriptomic findings validate the integrated network analysis, reinforcing TMEM189 as the primary functional candidate partner of TMEM45A for future mechanistic studies.

Functional Annotation of the TMEM45A–TMEM189-Associated Network

Functional enrichment analysis of the 150-gene TMEM45A-associated network across Gene Ontology and pathway databases identified significant enrichment in skin barrier establishment, epidermis development, skin epidermis development, and intermediate filament organization (GO BP; **Figure 14A; Supplementary Table S5**); the cornified envelope, desmosome, intermediate filament cytoskeleton, and cell–cell junction (**GO CC; Figure 14B; Supplementary Table S6**); wide-pore channel activity, calcium ion binding, gap junction channel activity, and RAGE receptor binding (GO MF; **Figure 14C; Supplementary Table S7**); IL-17 signaling, p53 signaling, arachidonic acid metabolism, and cornified envelope formation (KEGG; **Figure 14D; Supplementary Table S8**); cornified envelope formation, keratinization, metal sequestration by antimicrobial proteins, and antimicrobial peptide pathways (Reactome; **Figure 14E; Supplementary Table S9**); and pancreatic cancer subtypes, hair follicle development, calcium regulation in cardiac cells, and lipid particle composition (WikiPathways; **Figure 14F; Supplementary Table S10**)—collectively implicating this molecular network in epithelial differentiation, junctional organization, inflammatory signaling, and tumor progression.

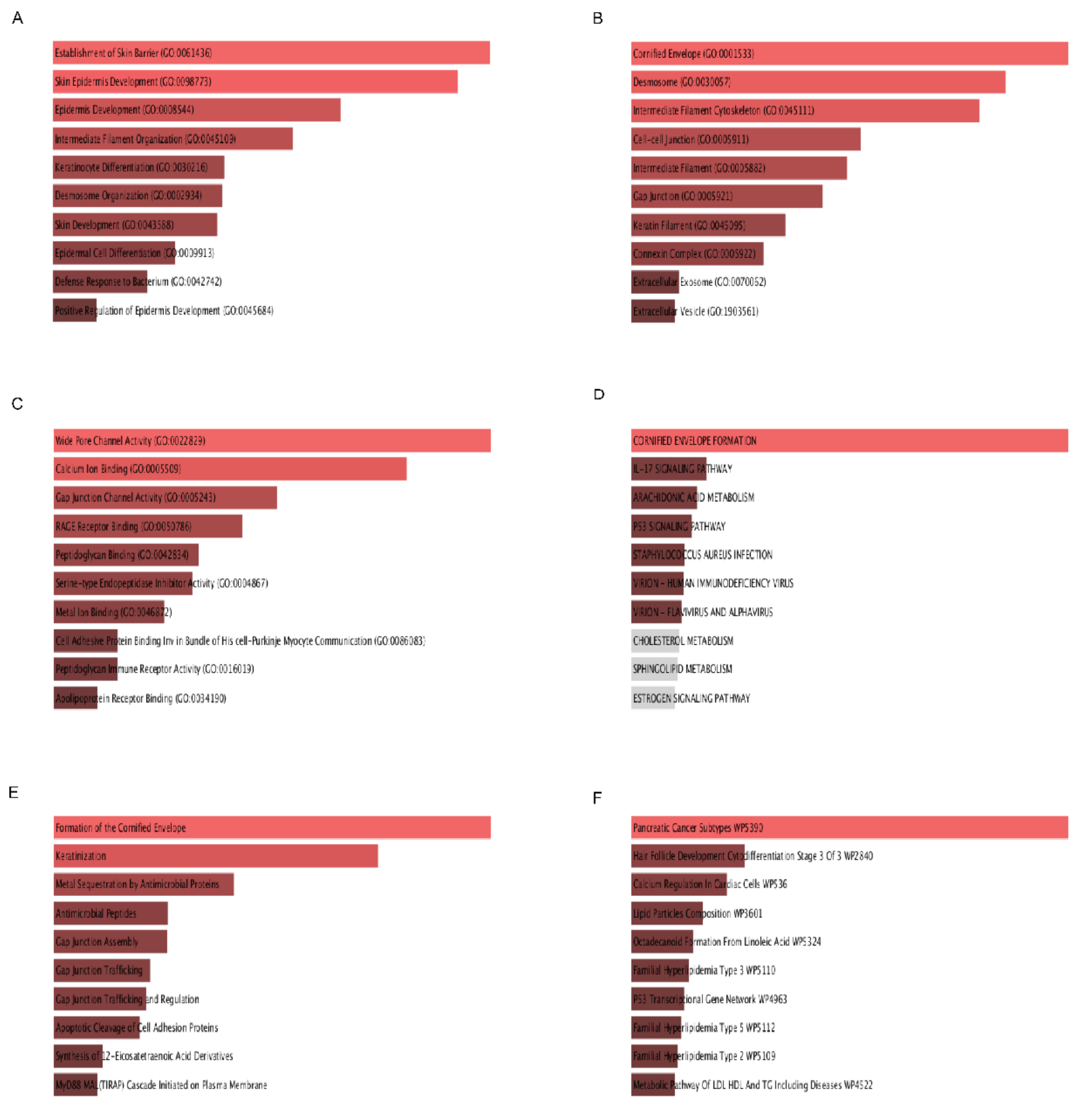

Figure 14: Pathway enrichment analysis and functional annotation of the molecular network linked to *TMEM45A*. Based on the 150 distinct genes that were found, the Enrichr platform was used to conduct the enrichment analysis. The top ten substantially enriched words in **(A)** Biological Process (BP), **(B)** Cellular Component (CC), and **(C)** Molecular Function (MF) are displayed in the Gene Ontology (GO) enrichment study. The findings emphasize important roles in channel activity, cornified envelope production, and skin barrier establishment. (D–F) Pathway enrichment study using three separate databases: **(D)** WikiPathways 2024, **(E)** Reactome 2024, and **(F)** KEGG 2026. The length of the bar and the color intensity (from red to light orange) match the significance level (p-value) in the bar charts that show the importance of enrichment.

Supplementary Tables S5–S10 provide detailed data such as odds ratios, adjusted p-values, and p-values.

Discussion

This study integrated transcriptomic, proteomic, epigenetic, genomic, immunological, and pharmacogenomic multi-omics analyses across 33 malignancies to characterize the oncogenic landscape and therapeutic relevance of TMEM45A. TMEM45A was overexpressed in multiple high-incidence solid tumors and consistently associated with adverse survival outcomes and recurrence, supporting its utility as a pan-cancer oncogenic biomarker (49–51). Integrated immune deconvolution and single-cell transcriptomics further linked elevated TMEM45A expression to an immune-excluded, myeloid-enriched, immunosuppressive tumor microenvironment with depleted cytotoxic lymphocyte infiltration(52,53). Collectively, these findings highlight that TMEM45A has a therapeutic target requiring experimental validation. TMEM45A was significantly overexpressed across aggressive malignancies—including BRCA, DLBC, GBM, HNSC, KIRC, LGG, LUSC, PAAD, and SARC—as corroborated by transcriptomic profiling (TIMER2.0, GEPIA2, GTEx) and protein-level evidence via HPA immunohistochemistry and CPTAC proteomics in BRCA, HNSC, and LIHC. Elevated TMEM45A correlated with advanced pathological stages in BRCA, KICH, KIRC, and LUSC, aligning with reports that TMEM45A associated with epithelial–mesenchymal transition (EMT), proliferation, and hypoxia adaptation in breast and ovarian cancers(54,55). These observations align with the established roles of oncogenic transmembrane proteins in driving tumor progression, invasion, and metastatic dissemination(56,57). Univariate Cox regression and Kaplan–Meier analyses consistently associated high TMEM45A expression with poor clinical outcomes, specifically predicting shorter OS and DFS in KIRC and STAD. Together, these findings support TMEM45A serve as a pan-cancer biomarker, aligning with the reported functions of other oncogenic transmembrane proteins (49,58).

SMART probe-level analysis identified significant hypomethylation at N-shore locus cg07907506 in BRCA, CHOL, KIRC, LIHC, LUAD, LUSC, PRAD, and UCEC, corroborated by CpG-aggregated analysis. UALCAN promoter analysis further confirmed significant hypomethylation across 11 TCGA cancer types, indicating promoter DNA hypomethylation as a key epigenetic factor of TMEM45A transcriptional activation (59,60). These findings are consistent with previous evidence indicating that epigenetic unmasking of N-shore regions facilitates transcription factor access to previously silenced loci during carcinogenesis(61,62). Genomic alteration analysis using cBioPortal demonstrated that copy-number amplification represented the predominant genomic alteration affecting TMEM45A, with the highest alteration frequencies observed in squamous malignancies, particularly LUSC (~8%) and CESC (~5.5%). Mutation mapping further identified a recurrent V236I substitution within the DUF716 domain. No statistically significant association was observed between TMEM45A genomic alterations and patient survival in high-alteration cohorts(63). Collectively, these findings suggest that the prognostic significance of TMEM45A is more closely associated with epigenetically regulated transcriptional activation than with recurrent somatic genomic alterations. These observations are consistent with the emerging concept that aberrant transcriptional activation, rather than recurrent genomic disruption alone, may contribute substantially to tumor progression across multiple cancer types. (64,65).

Beyond its cell-intrinsic oncogenic functions, computational immunogenomic analyses indicated that TMEM45A is associated with an immunosuppressive tumor microenvironment (TME) and distinct immune-cell infiltration patterns. TMEM45A expression was positively associated with StromalScore, ImmuneScore, and ESTIMATEScore, while EPIC and CIBERSORT analyses consistently demonstrated increased stromal infiltration and a myeloid-enriched immune landscape across multiple tumor types. TMEM45A expression was associated with a polarized immune landscape, showing negative correlations with cytotoxic CD8+ T cells and activated NK cells but positive correlations with cancer-associated fibroblasts (CAFs), M0/M1/M2 macrophages, neutrophils, and other myeloid cell populations. Single-cell transcriptomic analyses from TISCH2 further supported these observations by demonstrating cell type-specific TMEM45A expression patterns across the tumor microenvironment. TMEM45A expression was minimal in lymphoid populations but enriched in malignant epithelial cells, fibroblasts, and myeloid lineages, consistent with an extracellular matrix-rich, immune-excluded microenvironment that restricts lymphocyte infiltration(66,67). These observations are consistent with previous studies describing immune-excluded tumor microenvironments characterized by stromal remodeling and reduced cytotoxic lymphocyte infiltration(68,69). Moreover, the observed associations between TMEM45A expression and tumor mutation burden (TMB) or microsatellite instability (MSI), particularly in LAML, COAD, and MESO, suggest that TMEM45A may have potential utility in studies investigating immunotherapy responsiveness (70,71). Elevated TMEM45A expression was consistently associated with broad-spectrum drug resistance, particularly to the histone deacetylase (HDAC) inhibitor Vorinostat, across both the GDSC and CTRP datasets. The consistency of these findings across two independent pharmacogenomic datasets strengthens the evidence for an association between elevated TMEM45A expression and reduced sensitivity to HDAC inhibitors(72,73). Conversely, elevated TMEM45A expression was associated with increased sensitivity to Docetaxel, a microtubule-stabilizing agent, consistent with previous studies indicating that specific transmembrane proteins influence microtubule dynamics and enhance taxane sensitivity(74,75). Collectively, these findings support TMEM45A associated with pharmacogenomics that may assist future patient stratification according to therapeutic response, pending experimental and clinical validation.

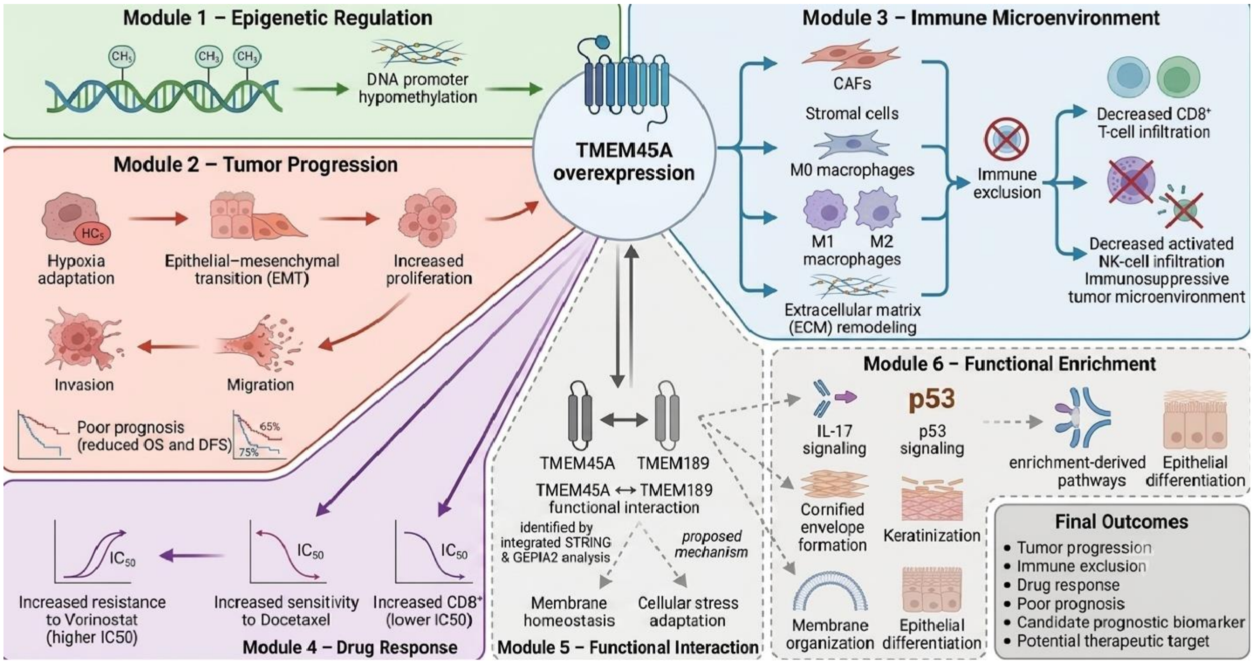

Figure 15: Mechanistic model of TMEM45A-mediated oncogenic progression across human cancers. Integrated multi-omics analyses suggest that promoter DNA hypomethylation contributes to TMEM45A overexpression, which is associated with multiple oncogenic processes, including hypoxia adaptation, epithelial–mesenchymal transition (EMT), enhanced proliferation, migration, invasion, and poor clinical outcomes. Elevated TMEM45A expression is further associated with an immune-excluded tumor microenvironment characterized by increased infiltration of cancer-associated fibroblasts (CAFs), stromal cells, and M0/M1/M2 macrophages, accompanied by reduced $CD8^+$ T-cell and activated natural killer (NK)-cell infiltration. Pharmacogenomic analyses indicate an association between high TMEM45A expression and reduced sensitivity to vorinostat but increased sensitivity to docetaxel. Integrated STRING protein–protein interaction and GEPIA2 co-expression analyses identified TMEM189 as the principal candidate functional partner of TMEM45A. Functional enrichment analyses further associated the TMEM45A-centered network with IL-17 signaling, p53 signaling, cornified envelope formation, keratinization, membrane organization, and epithelial differentiation. This schematic summarizes the proposed biological model generated from integrated transcriptomic, proteomic, epigenetic, immunological, pharmacogenomic, and single-cell analyses. Dashed arrows indicate inferred or enrichment-based associations that require experimental validation.

This study has several strengths, including its integrative multi-omics design and the systematic analysis of multiple complementary public cancer datasets. By integrating bulk transcriptomic data with single-cell RNA sequencing datasets from TISCH2, this study provides cell type-specific characterization of TMEM45A expression across epithelial, stromal, and myeloid compartments of the tumor microenvironment, thereby complementing bulk transcriptomic observations(76,77). Integration of transcriptomic findings with CPTAC proteomic data and Human Protein Atlas immunohistochemistry provided orthogonal validation of TMEM45A expression, increasing confidence in the observed expression patterns across selected cancer types(78). Another strength of this study is the identification of TMEM189 as the only overlapping gene between the STRING interaction network and GEPIA2 co-expression analysis, with consistent positive transcriptomic correlations across multiple TCGA cancer types. Given the reported roles of TMEM189 in plasmalogen biosynthesis and cellular stress responses, this association generates testable hypotheses regarding the potential involvement of the TMEM45A–TMEM189 axis in ferroptosis, lipid metabolism, and tumor progression(79,80). Finally, integration of independent pharmacogenomic datasets (GDSC and CTRP) identified consistent associations between elevated TMEM45A expression and reduced sensitivity to HDAC inhibitors, while supporting increased sensitivity to docetaxel, thereby enhancing the robustness of the pharmacogenomic findings.

This study has several limitations. First, exclusive reliance on public TCGA and GTEx datasets introduces potential selection and batch-related biases due to heterogeneous sample acquisition, processing pipelines, and patient cohorts. Second, the scarcity of matched normal tissue controls in several cancer types constrained comprehensive validation across all malignancies. Third, despite consistent TMEM45A–TMEM189 transcriptomic co-expression, the mechanistic basis of this interaction and its associated pathways (e.g., p53 and IL-17 signaling) remain unvalidated. Finally, as an entirely computational study, in vitro and in vivo studies are required to confirm whether TMEM45A directly regulates epithelial–mesenchymal transition (EMT), macrophage polarization, immune exclusion, and HDAC inhibitor resistance.

Conclusion

This pan-cancer analysis identifies TMEM45A, whose overexpression associates with promoter hypomethylation, adverse prognosis, an immune-excluded and myeloid-enriched tumor microenvironment, and distinct pharmacogenomic profiles (e.g., predicted docetaxel sensitivity and vorinostat resistance). Furthermore, consistent TMEM45A–TMEM189 transcriptomic co-expression across TCGA cohorts highlights a functional axis requiring future in vitro and in vivo validation to clarify its role in immune exclusion, therapeutic response, and tumor progression.

Declarations

Transparency Statement

The author certifies that this study was conducted and reported with integrity and transparency. All multi-omics datasets (including TCGA, GTEx, CPTAC, HPA, GDSC, CTRP, and TISCH2), analytical workflows, and bioinformatics procedures are accurately described, and no relevant data or analyses have been intentionally omitted.

Ethics Approval and Consent to Participate

Ethics approval and informed consent were not required because this study exclusively analyzed publicly available, de-identified datasets and did not involve direct participation of human subjects or animals.

Consent for Publication

Not applicable. This study contains no identifiable personal information, clinical images, or individual patient data requiring consent for publication.

Data Availability Statement

All datasets analyzed during this study are publicly available. Transcriptomic, survival, immune, and methylation analyses were conducted using publicly available resources, including TIMER2.0, GEPIA2, SMART, UALCAN, and the TCGAplot R package. Proteomic and immunohistochemical data were obtained from the Clinical Proteomic Tumor Analysis Consortium (CPTAC) and the Human Protein Atlas (HPA). Genomic alteration data were retrieved from cBioPortal for Cancer Genomics. Single-cell RNA sequencing datasets were obtained from TISCH2, whereas pharmacogenomic data were retrieved from the Genomics of Drug Sensitivity in Cancer (GDSC) and Cancer Therapeutics Response Portal (CTRP). All corresponding web resources and accession details are provided in the Methods section.

Code Availability

The custom R scripts, bioinformatics pipelines, and analysis workflow supporting this study are publicly available at: https://github.com/shakawat96907/Pan-Cancer-Analysis-TMEM45A.

Competing Interests

The author declares no competing interests.

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Author Contributions

Md. Shakawat Hossain conceived and designed the study; performed data curation, bioinformatics, and statistical analyses; interpreted the results; generated the figures and visualizations; drafted and critically revised the manuscript; and approved the final version of the manuscript.

Declaration of Generative AI and AI-Assisted Technologies

The author used generative AI tools solely to improve the language, grammar, clarity, and organization of the manuscript. Gemini, Grammarly, Paperpal, and QuillBot were used exclusively for language editing and stylistic refinement. Consensus AI was used to assist in identifying relevant scientific literature. All scientific analyses, data interpretation, conclusions, and editorial decisions were performed, critically verified, and approved by the author, who accepts full responsibility for the content of this manuscript.

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Supplementary Information

Supplementary materials are provided as two separate files: **Supplementary Figures** (Figures S1–S5 with corresponding legends) and **Supplementary Tables** (Tables S1–S10), which provide additional data supporting the findings of this study.

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